RUTH ELVIN HARIANJA

Data Scientist | Machine Learning Specialist

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EDUCATION

UNIVERSITAS SUMATERA UTARA (USU) **Medan, Indonesia**

Master of Science in Data Science & Artificial Intelligence (S2 DSAI) *2024 – 2026*

- Thesis & Research Focus: Geospatial Anomaly Detection, Spatio-Temporal Clustering (ST-DBSCAN), and Explainable AI (SHAP).

INSTITUT TEKNOLOGI DEL **Toba, Indonesia**

Bachelor of Applied Science in Software Engineering (D-IV RPL) *2017 – 2021*

- Graduated with comprehensive coursework in Algorithms, Data Structures, Database Systems, and Software Architecture.

WORK EXPERIENCE

City Government of Medan **Medan, Indonesia**

Data Science & Machine Learning Intern

- Developed an interactive 3D analytics dashboard (Streamlit, ST-DBSCAN, PyDeck 3D) implementing spatio-temporal clustering and Haversine distance to detect attendance fraud and spatial deviations among civil servants.
- Engineered a multi-factor weighted risk scoring engine (stratified into LOW, MEDIUM, HIGH risk) to automate compliance audits and inform executive administrative decisions.

Gokomodo **South Jakarta, Indonesia**

Quality Assurance Engineer

- Led end-to-end quality assurance, test scenario design, cross-functional Agile collaboration, and backend database validation for enterprise B2B e-commerce platforms.
- Formulated complex SQL validation queries and API integration tests to ensure data integrity across transactional pipelines.

Anteraja **South Jakarta, Indonesia**

Software Quality Assurance

- Executed functional, integration, regression, and data verification testing for high-volume logistics and package tracking workflows using JIRA, TestRail, and database auditing.

Transfree **South Jakarta, Indonesia**

Software Quality Assurance Intern

- Assisted in test plan design, test execution, API verification, and UAT for cross-border financial transaction and remittance systems.

RELEVANT DATA SCIENCE & AI PROJECTS

Attendance Anomaly Web App

Technologies: Python, Flask, PyOD (Isolation Forest, LOF, ECOD), SHAP XAI, Leaflet.js, Plotly

- Built an end-to-end production Flask web application deploying an ensemble of Isolation Forest, LOF, and ECOD for multi-algorithm outlier detection.
- Integrated SHAP (XAI) for transparent feature attribution, automated data preprocessing, and Leaflet.js interactive geospatial mapping with automated Excel reporting.

Beverages Sales Time Series Forecasting Engine

Technologies: TensorFlow, Keras LSTM, Statsmodels ARIMA, Scikit-Learn, Pandas

- Modeled 55k+ retail transactions comparing ARIMA baseline against Multi-layer LSTM Neural Networks to forecast demand volume and capture non-linear trends.
- Evaluated exogenous promotional drivers (onpromotion) and commodity prices, demonstrating significant forecast accuracy improvements over baseline models.

Credit Card Fraud Detection & Behavioral Feature Pipeline

Technologies: Scikit-Learn, Random Forest, Imbalanced Learning, Feature Engineering

- Constructed a high-precision fraud detection engine on extreme imbalanced data (0.58% fraud rate) utilizing Random Forest with cost-sensitive class balancing.
- Engineered behavioral domain features including transaction velocity (velocity_1h), time deltas, rolling expenditure averages, and Haversine distance.

TECHNICAL SKILLS

- **Machine Learning & AI:** Scikit-Learn, TensorFlow/Keras, PyOD (Isolation Forest, LOF, ECOD), Random Forest, SHAP (XAI), Imbalanced Classification, Genetic Algorithms.
- **Data Science & Analytics:** Python (Pandas, NumPy, Scipy), SQL (PostgreSQL, MySQL), Time Series Forecasting (LSTM, ARIMA), Feature Engineering, Statistical Analysis, EDA.
- **Geospatial & Web Tools:** ST-DBSCAN, Spatial DBSCAN, Haversine Distance, Flask, Streamlit, PyDeck 3D, Leaflet.js, Git/GitHub, JIRA, Postman.